



Disturbance rejection of ball mill grinding circuits using DOB and MPC

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ABSTRACT

Ball mill grinding circuit is essentially a multivariable system with couplings, time delays and strong disturbances. Many advanced control schemes, including model predictive control (MPC), adaptive control, neuro-control, robust control, optimal control, etc., have been reported in the field of grinding process. However, these control schemes including the MPC scheme usually cannot achieve satisfying effects in the presence of strong disturbances. In this paper, disturbance observer (DOB), which is widely used in motion control applications, is introduced to estimate the disturbances in grinding circuit. A compound control scheme, consisting of a feedforward compensation part based on DOB and a feedback regulation part based on MPC (DOB-MPC), is thus developed. A rigorous analysis of disturbance rejection performance is given with the considerations of both model mismatches and external disturbances. Simulation results demonstrate that when controlling the ball mill grinding circuit, the DOB-MPC method possesses a better performance in disturbance rejection than that of the MPC method.

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1. Introduction

Ball mill grinding circuits are among the most important operation units in mineral processing plants. Grinding processes hold almost 50% of the total expenditure of the concentrator plant. Moreover, the product particle size influences the recovery rate of the valuable minerals as well as the volume of tailing discharge in the subsequent processes. Generally speaking, the aim of grinding processes is to maintain stable and effective control of the product particle size and the circulating load. However, undesirable characteristics, such as strong interactions between the variables, complex dynamic features, time delays, and strong disturbances, always bring about difficulties and problems to both control engineers and researchers. To this end, besides traditional PID controllers [1], many advanced approaches, such as model predictive control (MPC) [1–5], adaptive control [6,7], neuro-control [8,9], robust control [10,11], optimal control [12], supervisory control [13] and expert control [14,15], etc., are reported in recent years.

As mentioned above, MPC has been researched and applied as one of the most popular advanced control schemes in grinding field. The reason may lie in that MPC possesses the following remarkable advantages: (1) capability of handling multi-input-multi-output (MIMO) processes, (2) ability to control difficult processes, such as processes with time delay part and non-minimum-phase part, (3) capability of handling constraints of both manipulated and controlled variables, and (4) simplicity of modeling (generally depending on step/impulse response), etc.

However, in ball mill grinding circuit, the control performance of the closed-loop system is severely interrupted by various disturbances which are generally complex or unmeasurable. Usually, these disturbances can be divided into external and internal ones. The external disturbances often include the variations of the ore hardness and feed particle size, and so on, while the internal disturbances are generally caused by model mismatches or coupling effects. Variations of the ore hardness and feed particle size may cause continuous fluctuation of the product particle size. Model mismatches and coupling effects may lead to adverse dynamic performance and even result in unstable control of the closed-loop system.

It should be pointed out that MPC, as well as many other advanced control schemes, rejects disturbances merely through the action of feedback regulation part and does not deal with the disturbances directly by controller design. When meeting strong disturbances, these methods may lead to some limitations, for example, the dynamic process of the closed-loop system may become sluggish and the closed-loop system may even become unstable due to the reason that the fluctuations of the controlled variables may become too large.

In order to improve the disturbance rejection performance of the grinding circuits, a feedforward compensation part for the disturbances is introduced to the controller besides a conventional feedback part. However, usually, it is hard or even impossible to measure the disturbances of grinding circuits. A feasible solution is to develop disturbance estimation technique. Disturbance observer (DOB) is known as an effective technique to estimate disturbances and has been extensively applied for feedforward compensation design in the presence of disturbances. The DOB technique was originally proposed in the field of motion control by Ohishi et al. [16]. Disturbance observer based control schemes for linear and nonlinear systems have been

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proposed and applied successively in many other control fields, e.g., robotic systems [17–19], hard disk drive systems [20–22], position system [23], missile system [24] and general systems [25–27], etc. The successful applications of DOB may attribute to its remarkable features: (1) the design of DOB usually does not rely on precise disturbance models, (2) DOB can be applied to handle the unmeasurable disturbances, and (3) DOB handles the disturbances without distinguishing external and internal ones.

In this paper, a compound control scheme combining a feedforward compensation part based on DOBs and a feedback regulation part based on MPC (DOB-MPC) is developed to improve the disturbance rejection performance of the closed-loop grinding system. One novelty of this work lies in that the terms of model mismatches in the closed-loop system are merged into the disturbance term and are regarded as a part of the lumped disturbances, in other words, the disturbance rejection analysis here considers not only the external disturbances, but also the effects caused by model mismatches. The other novelty lies in that a multivariable compound control structure based on DOB and MPC is proposed, thus the DOB-MPC scheme inherits many advantages of both MPC and DOB, some of which are listed as follows:

- Capability in handling processes with time delays;
- Ability to dealing with multivariable systems (good decoupling property); and
- Remarkable superiorities in handling strong disturbances (including both external disturbances and internal disturbances caused by model mismatches).

The rest of the paper is organized as follows. The process background and features of the grinding circuits are illustrated in Section 2. In Section 3, MPC algorithm and DOB technique are introduced at first, and then a compound DOB-MPC scheme is elaborated and applied to control the ball mill grinding circuit. Section 4 carries out simulation studies, and the disturbance rejection performance of the proposed DOB-MPC scheme is compared with that of the MPC scheme. Finally, conclusions are given in Sections 5.

2. Process description

2.1. Process background

Raw ores cannot be used as final products for industrial uses. The aim of mineral processing is to concentrate the valuable minerals for the

subsequent extraction processing [28]. The valuable minerals are originally contained in an oxide or sulfide. It is a long and difficult process to liberate the valuable minerals from the coarse ores. Usually, the overall mineral processing can be divided into the following three parts:

- Ore size reduction process (crushing, grinding and classifying).
- Separation (flotation, magnetic or gravimetric separation) and concentration process.
- Tailing discharge.

The raw ores cannot be sent to the ball mill directly because the size is generally too large for grinding. The crushing process, aiming to reduce the size of the raw ore, supplies the appropriate size of feed ore for the grinding process. The quality of the product particle will affect the subsequent separation process. Moreover, the efficiency of this process has tight relation with the overall economic benefits of the concentrator plant. According to the ore property, specific methods can be chosen for separation, such as flotation, magnetic and gravimetric separation, etc. Finally, tailings disposal is of great importance from the viewpoint of environment protection.

2.2. Description of grinding circuit

Ball mill grinding circuit is one of the most important mineral processing units in concentrator plants. A typical ball mill grinding circuit is shown in Fig. 1, which consists of a ball mill, a pump sump, hydro-cyclones and solids feeding conveyors.

During the production process, the coarse ore is fed into the mill together with the mill feed water. The tumbling action of the balls within the revolving mill crushes the feed ore to finer sizes. The slurry with finer particle is discharged from the mill to a pump sump. Such slurry is diluted in the pump sump by adding dilution water, and then pumped to hydro-cyclones for classification. After the classification process, the slurry is separated into two streams: an overflow stream containing the finer particles which is regarded as the desired product and an underflow stream containing the larger particles as the circulating load which is recycled back to the ball mill for regrinding.

Product particle size y_1 (%) is generally viewed as the most important controlled variable in a grinding circuit. Mine ore is required to be ground to a specified size called the liberation size that the imbedded mineral particles are exposed for effective recovery in the downstream processing. Circulating load y_2 (t/h) is known as

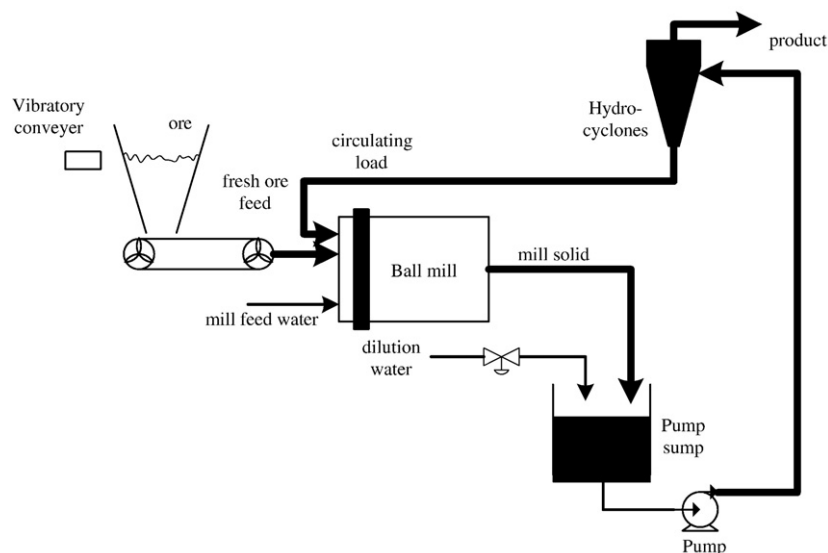


Fig. 1. Schematic diagram of a ball mill grinding circuit.

another important controlled variable. Larger or smaller circulating load than the desirable setpoint may degrade the efficiency of the whole circuit. Note that mill solids concentration (%solids) and sump level (m) are another two important variables needed to be controlled. In real practice, two local controllers are usually designed for such two controlled variables: the sump level is maintained within a limited range by regulating the slurry pumping rate (Hz), while the mill solids concentration is kept around a desired setpoint by regulating the mill addition water (m³/h) in (constant or variable) proportion to the fresh ore feed rate [3,13]. Control of the sump level and mill solids concentration is therefore omitted here for space, and only two controlled variables, viz the product particle size y_1 and circulating load y_2 are considered.

Strong external disturbances in grinding circuits, such as variations of ore hardness d_{oh} and feed particle size d_{fps} , may bring about some undesirable dynamics to the controlled variables, e.g., the product particle size may fluctuate continuously in such case. Both fresh ore feed rate u_1 (t/h) and dilution water flow rate u_2 (m³/h) can be manipulated to regulate the two selected controlled variables [13]. The relationships among the selected controlled variables, manipulated variables and external disturbances in grinding process are illustrated in Fig. 2.

3. Control scheme

In this work, a compound control scheme is proposed to solve difficult control problems in grinding circuit. Firstly, multivariable MPC algorithm and DOB technique are introduced. A DOB-MPC control scheme is then illustrated in details.

3.1. Multivariable MPC algorithm

Multivariable MPC algorithm was originally proposed by Cutler et al. [29] and discussed extensively by many researchers. Here it is summarized for the convenience of the reader. Considering a multivariable system with m manipulated variables and p controlled variables, the quadratic performance objective function of an unconstrained MPC algorithm possesses the following form

$$\min J(k) = [y_r(k) - \tilde{y}_p(k)]^T \mathbf{Q} [y_r(k) - \tilde{y}_p(k)] + [\Delta u_M(k)]^T \mathbf{R} [\Delta u_M(k)], \quad (1)$$

where $y_r(k)$ stands for the setpoints of p controlled variables in the future P sampling instances, and $\tilde{y}_p(k)$ for the predicted outputs of p controlled variables in the future P sampling instances. $\Delta u_M(k)$ represents the changes of m manipulated variables in the future M sampling instances. P and M represent the prediction horizon and

control horizon, respectively. $\mathbf{Q}$ ($Pp \times Pp$) and $\mathbf{R}$ ($Mm \times Mm$) represent the error weighting matrix and input weighting matrix, respectively. The model predictions of the outputs are stated as

$$\tilde{\mathbf{y}}_N(k) = \tilde{\mathbf{y}}_0(k) + \mathbf{A} \Delta u_M(k), \quad (2)$$

where $\tilde{\mathbf{y}}_N(k)$ and $\tilde{\mathbf{y}}_0(k)$ are the model prediction vector and the initial predicted output vector without control action, respectively. $\mathbf{A}$ ($Pp \times Mm$) is a multivariable MPC dynamic matrix formed from the unit step response coefficients. Due to the effects of measurement noise, external disturbances and model mismatches, the predicted outputs are generally not accurate and should be corrected by the real outputs $y(k+1) = [y_1(k+1), \dots, y_p(k+1)]^T$ to realize closed-loop control. Construct $\mathbf{e}(k+1)$ as the vector of predicted errors for the p controlled variables

$$\begin{aligned} \mathbf{e}(k+1) &= [\mathbf{e}_1(k+1), \dots, \mathbf{e}_p(k+1)]^T \\ &= [y_1(k+1) - \tilde{y}_{1,1}(k+1|k), \dots, y_p(k+1) - \tilde{y}_{p,1}(k+1|k)]^T, \end{aligned} \quad (3)$$

where $\tilde{y}_{i,1}(k+1|k)$, $i = 1, \dots, p$, is the model prediction of p controlled variables at the $(k+1)$ th sampling instance. Thus the predicted outputs after correction can be represented as

$$\tilde{\mathbf{y}}_p(k) = \tilde{\mathbf{y}}_N(k) + \mathbf{H} \mathbf{e}(k+1), \quad (4)$$

where $\tilde{\mathbf{y}}_p(k)$ is the predictive output vector after correction, $\mathbf{H}$ is the corrected coefficient matrix. It can be solved from the multivariable MPC performance objective function presented in Eq. (1) that

$$\Delta u_M(k) = (\mathbf{A}^T \mathbf{Q} \mathbf{A} + \mathbf{R})^{-1} \mathbf{A}^T \mathbf{Q} [y_r(k) - \tilde{\mathbf{y}}_p(k)]. \quad (5)$$

MPC scheme is implemented in a receding horizon framework. At any sampling instance, the optimization problem is formulated over the prediction horizon and an optimized manipulated variable trajectory is solved from the objective function. Only the first move is applied to the plant and this step is repeated for the next sampling instance.

3.2. Disturbance observer for process with time delays

In grinding process, the conventional disturbance observer is no longer available due to the reason that the inverse function of time delay part is physical infeasible. To this end, a modified disturbance observer with the considerations of time delays [30] is adopted here. Consider a two-input–two-output process with two external disturbances which is expressed as

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} + \begin{bmatrix} H_{11}(s) & H_{12}(s) \\ H_{21}(s) & H_{22}(s) \end{bmatrix} \begin{bmatrix} D_{oh}(s) \\ D_{fps}(s) \end{bmatrix}, \quad (6)$$

with

$$P_{ii}(s) = p_{ii}(s) e^{-\tau_{ii}s} (i = 1, 2), \quad (7)$$

$$D_{exi}(s) = H_{i1}(s) D_{oh}(s) + H_{i2}(s) D_{fps}(s) (i = 1, 2), \quad (8)$$

where $U_i(s)$ ($i = 1, 2$) the manipulated variables, $Y_i(s)$ ($i = 1, 2$) the controlled variables, $D_{oh}(s)$ and $D_{fps}(s)$ the external disturbances, $D_{exi}(s)$ ($i = 1, 2$) the effects of external disturbances on $Y_i(s)$, $P_{ij}(s)$ ($i = 1, 2; j = 1, 2$) the models of the process channels, $p_{ii}(s)$ ($i = 1, 2$) the minimum-phase part of $P_{ii}(s)$, and $H_{ij}(s)$ ($i = 1, 2; j = 1, 2$) the models of the disturbance channels.

Assume that u_1 is manipulated to control y_1 , also u_2 is manipulated to control y_2 . The structure of the modified disturbance observer for

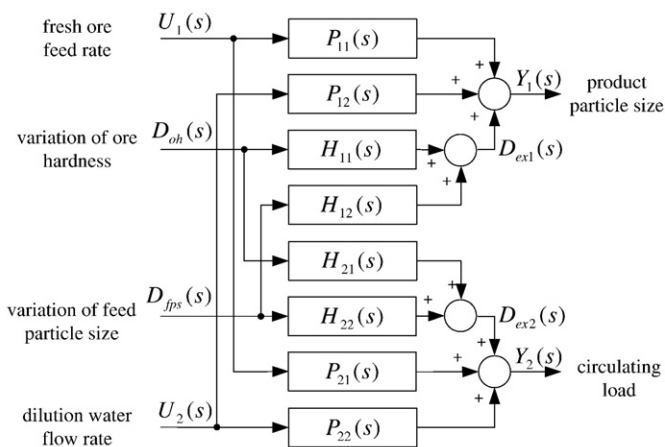


Fig. 2. The relationships among the variables in grinding circuit.

$u_1 - y_1$ loop is shown in Fig. 3(a). Note that the case of $u_2 - y_2$ loop is similar which is omitted here. Clearly, the model $p_{11}(s)$ is separated into a multiplication of a minimum-phase part p_{11} and a time delay part $e^{-\tau_{11}s}$ as below

$$P_{11}(s) = p_{11}(s)e^{-\tau_{11}s}. \quad (9)$$

Similarly, the nominal model of the $u_1 - y_1$ loop, i.e., $G_{11}(s)$ can be represented as

$$G_{11}(s) = g_{11}(s)e^{-\tau_{11}s}. \quad (10)$$

Considering the effect of model mismatches, an equivalent block diagram is obtained, as shown in Fig. 3(b). The final form of the modified

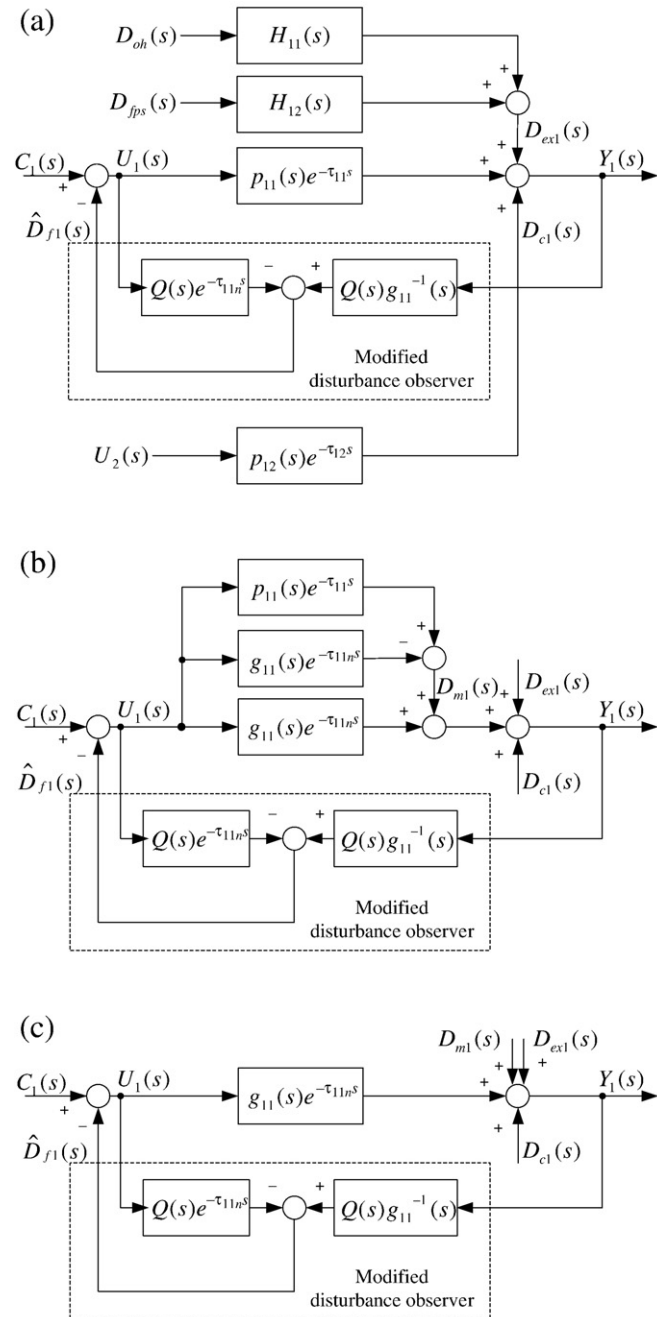


Fig. 3. Block diagram of the modified disturbance observer for process with time delays: (a) an initial form; (b) an equivalent form; and (c) the final form.

disturbance observer is shown in Fig. 3(c). Clearly, the lumped disturbance $D_1(s)$ of $u_1 - y_1$ loop includes the external disturbance $D_{ex1}(s)$ and the internal disturbances consisting of $D_{m1}(s)$ caused by model mismatches and $D_{c1}(s)$ caused by coupling effects, i.e.,

$$D_1(s) = D_{ex1}(s) + D_{m1}(s) + D_{c1}(s), \quad (11)$$

where

$$D_{m1}(s) = [p_{11}(s)e^{-\tau_{11}s} - g_{11}(s)e^{-\tau_{11}s}]U_1(s), \quad (12)$$

$$D_{c1}(s) = p_{12}(s)e^{-\tau_{12}s}U_2(s). \quad (13)$$

One obtains from Fig. 3(c) that

$$Y_1(s) = g_{11}(s)e^{-\tau_{11}s}U_1(s) + D_{m1}(s) + D_{ex1}(s) + D_{c1}(s). \quad (14)$$

Substituting Eq. (11) into Eq. (14), yields

$$Y_1(s) = g_{11}(s)e^{-\tau_{11}s}U_1(s) + D_1(s). \quad (15)$$

It can be obtained from Fig. 3(c) that

$$\hat{D}_{f1}(s) = Q(s)g_{11}^{-1}(s)Y_1(s) - Q(s)e^{-\tau_{11}s}U_1(s). \quad (16)$$

Substituting Eq. (15) into Eq. (16), yields

$$\hat{D}_{f1}(s) = Q(s)g_{11}^{-1}(s)D_1(s). \quad (17)$$

Define $\tilde{D}_i(s)$ ($i = 1, 2$) as the error between the real value and the estimation value of the lumped disturbance in $u_i - y_i$ loop, i.e.,

$$\tilde{D}_1(s) = D_1(s) - g_{11}(s)e^{-\tau_{11}s}\hat{D}_{f1}(s). \quad (18)$$

Substituting Eq. (17) into Eq. (18), yields

$$\tilde{D}_1(s) = [1 - Q(s)e^{-\tau_{11}s}]D_1(s). \quad (19)$$

According to the Final-value Theorem, one obtains from Eq. (19) that

$$\begin{aligned} \tilde{d}_1(\infty) &= \lim_{s \rightarrow 0} s\tilde{D}_1(s) \\ &= \lim_{s \rightarrow 0} [1 - Q(s)e^{-\tau_{11}s}] \lim_{s \rightarrow 0} sD_1(s) \\ &= \lim_{s \rightarrow 0} [1 - Q(s)e^{-\tau_{11}s}]d_1(\infty). \end{aligned} \quad (20)$$

Obviously, if selecting $Q(s)$ such that the steady-state gain of $Q(s)$ is 1, then it can be obtained that

$$\tilde{d}_1(\infty) = 0. \quad (21)$$

It can be found from Eq. (21) that the disturbances can be asymptotically rejected. Hence, the modified disturbance observer is a practical approach to deal with the process with time delays.

3.3. DOB-MPC scheme for ball mill grinding circuits

The method of this work focuses on disturbance rejection, including model mismatches as well as external disturbances. A DOB-MPC scheme is proposed to control the ball mill grinding circuits in this part. The detailed design procedures are described as follows.

Consider a typical two-input–two-output grinding circuit shown in Fig. 1. The manipulated variables are fresh ore feed rate u_1 (t/h) and dilution water flow rate u_2 (m³/h), while the controlled variables are product particle size y_1 (%–200 mesh) and circulating load y_2 (t/h), respectively. According to the requirements of the real practice, the

constraints of the manipulated and controlled variables are stated as follows: (1) the minimum and maximum values of the fresh ore feed rate are 60 (t/h) and 70 (t/h), (2) the minimum and maximum values of the dilution water flow rate are 40 (m³/h) and 50 (m³/h), (3) the minimum and maximum values of the product particle size are 68 (%–200 mesh) and 72 (%–200 mesh), and (4) the minimum and maximum values of the circulating load are 140 (t/h) and 170 (t/h). Assume that the fresh ore feed rate u_1 is manipulated to control the product particle size y_1 , and the dilution water flow rate u_2 is manipulated to control the circulating load y_2 . In this case, the schematic block diagram of the DOB-MPC scheme for the ball mill grinding circuit is shown in Fig. 4.

The internal structure of the feedforward-feedback DOB-MPC controller is shown in Fig. 5. It can be observed that a MPC controller and two disturbance observers are employed here to construct the compound DOB-MPC structure. Fine effects can be achieved by tuning the parameters of MPC controller and the time constants of the filter $Q(s)$ in DOB. Here $Q(s)$ is selected as a first-order-low-pass filter with the steady-state gain of 1, which can be expressed as

$$Q(s) = \frac{1}{\lambda s + 1}, \lambda > 0. \quad (22)$$

It should be pointed out that the implementation of DOB is rather simple, thus the introduction of feedforward compensation part does not increase much computational complexity.

4. Performance analysis and comparisons

For control study, plant test has been constructed to develop the input-output transfer function model of the grinding circuit as follows:

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}, \quad (23)$$

where

$$G_{11}(s) = g_{11}(s)e^{-\tau_{11}s} = \frac{-0.58}{2.5s + 1}e^{-0.68s}, \quad (24)$$

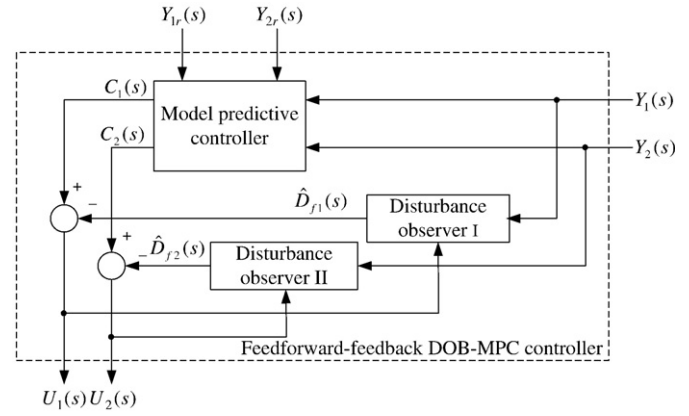


Fig. 5. The internal structure of the feedforward-feedback DOB-MPC controller.

$$G_{12}(s) = g_{12}(s)e^{-\tau_{12}s} = \frac{4(1-0.9938e^{-0.47s})}{(2s+1)(6s+1)}e^{-0.2s}, \quad (25)$$

$$G_{21}(s) = g_{21}(s)e^{-\tau_{21}s} = \frac{2.2}{6s+1}e^{-0.6s}, \quad (26)$$

$$G_{22}(s) = g_{22}(s)e^{-\tau_{22}s} = \frac{2.83}{3.5s+1}e^{-0.13s}. \quad (27)$$

Note that the time constants are expressed in minutes in this paper. As for such nominal model, a feasible variable pairing is $u_1 - y_1$ and $u_2 - y_2$ according to relative gain array (RGA) analysis. In this section, the disturbance rejection performance is studied in the nominal case as well as the model mismatch case. Here the external disturbances are imposed on the process through disturbance channels. According to Ref. [31], ore hardness and feed particle size have great influence on the product particle size and the circulating load. For example, ore feed with increased ore hardness leads to a decreased particle size and an increased circulating load, and these dynamics can be modeled as a first-order plus dead-time (FOPDT) form which is the most commonly used model to describe the dynamics of industrial process. The grinding process can be described

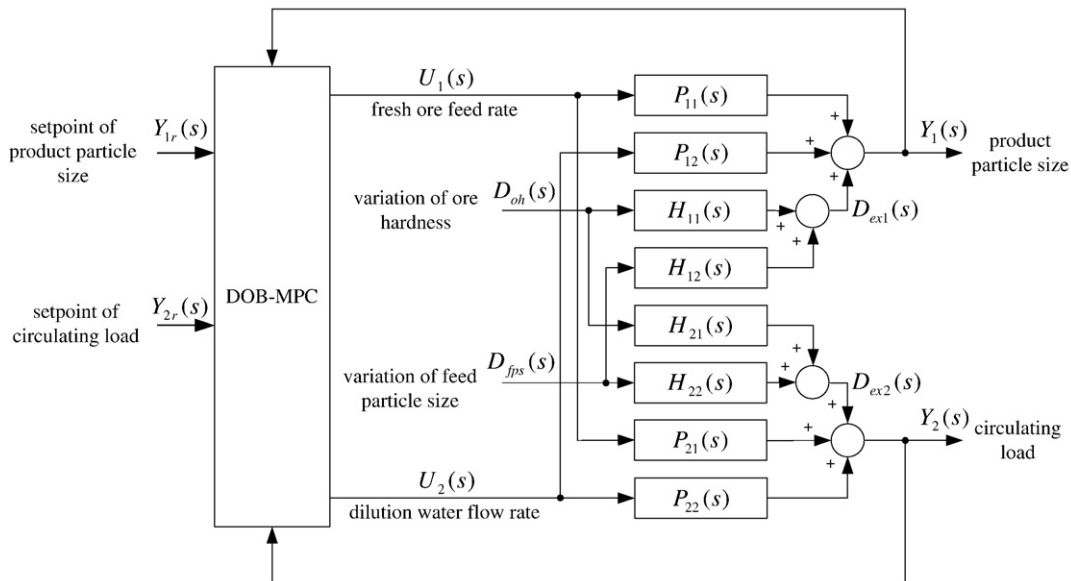


Fig. 4. The schematic block diagram of the feedforward-feedback DOB-MPC scheme for grinding circuit.

Table 1
Tuning parameters of the controllers for grinding process.

Method	$T_s(\text{min})$	P	M	Q	R	$\lambda_1(\text{min})$	$\lambda_2(\text{min})$
MPC	0.5	40	10	diag [1.0 0.4]	diag [0.5 0.5]	–	–
Proposed	0.5	40	10	diag [0.2 0.2]	diag [1.0 1.0]	0.1	1.5

as Eq. (6). Assume that the transfer functions of the disturbance channels are expressed as follows for simulation study

$$H_{11}(s) = \frac{-6.4}{4.8s + 1} e^{-0.52s}, \quad (28)$$

$$H_{12}(s) = \frac{-8.6}{5.2s + 1} e^{-0.58s}, \quad (29)$$

$$H_{21}(s) = \frac{32}{5.3s + 1} e^{-0.46s}, \quad (30)$$

$$H_{22}(s) = \frac{40}{6.2s + 1} e^{-0.63s}, \quad (31)$$

In Eq. (6), $D_{oh}(s)$ and $D_{fps}(s)$ represent variations of ore hardness and feed particle size, respectively. It should be pointed out that both ore hardness and the feed particle size are usually hard or even impossible to be measured. Thus here they are expressed in a scaled form rather than a real physical unit form. For example, $D_{oh}(s) = 10\%$ means that the ore hardness has an increase of 10% compared with its nominal value.

The disturbance rejection performance under the control of the proposed method is compared with that under the MPC method. The tuning parameters of the two controllers are listed in Table 1.

4.1. Disturbance rejection in the nominal case

Since here we study the external disturbance rejection performance of the grinding process under the control of the proposed method, the nominal case is considered, i.e., the transfer function models of the process channels satisfy that $P_{ij}(s) = G_{ij}(s)$ ($i = 1, 2$; $j = 1, 2$). In this part, two kinds of simulations have been carried out to demonstrate the external disturbance rejection property of the proposed method. Generally, the ore hardness and feed particle size fluctuates continuously during the whole process of production [31], thus a sinusoidal disturbance is likely to match the features of the real practice better than a common step disturbance does. To this end,

besides the step disturbance case, a sinusoidal disturbance case is also considered and simulated.

Case I. Step external disturbances in the nominal case: the ore hardness has an increase of 20% at $t = 20$ min, while the feed particle size has a decrease of 20% at $t = 100$ min.

Fig. 6 shows the response curves of the controlled variables under the control of both DOB-MPC (denoted by solid line) and MPC (shown by dash-dot line) in Case I. The corresponding changes of the manipulated variables are shown in Fig. 7. The effects of the external disturbances and the estimations on the controlled variables are shown in Fig. 8. It can be observed from Fig. 6 that the dynamic performances of both the product particle size and the circulating load under the proposed method (solid line) are much better than those under the MPC method (dash-dot line). Compared with the MPC scheme, the proposed method obtains a faster convergence speed, smaller amplitudes of fluctuations and shorter settling times, etc. As shown in Fig. 8, the error between the estimated (solid line) and real (dotted line) disturbances are very small, which means that the disturbance observer can effectively estimate the effects caused by external disturbances.

To quantitatively analyze the disturbance rejection performance, two performance indexes including peak overshoot and integral of absolute error (IAE) are employed. The performance indexes in such case are shown in Table 2. It can be observed from Table 2 that the overshoots for both the product particle size and the circulating load under the proposed method are much smaller than those under the MPC method. The IAE values of both the controlled variables under the proposed method are also much smaller than those under the MPC scheme.

Case II. Sinusoidal external disturbances in the nominal case: variation of the ore hardness is sinusoidal, i.e., $d_{oh} = A \sin(\omega t)$, here $A = 20\%$, $\omega = 0.1$ rad/min.

Fig. 9 shows the response curves of the controlled variables under the control of the two methods in the case of a sinusoidal disturbance. The corresponding variations of the manipulated variables are shown in Fig. 10. Also the effects of the real disturbances and the estimations by disturbance observer are shown in Fig. 11.

It can be observed from Fig. 9 that the fluctuation amplitudes of both the product particle size and the circulation load under the proposed method (solid line) are much smaller than those under the MPC method (dash-dot line), although their fluctuation frequencies under the two control methods are almost the same. Fig. 10 shows

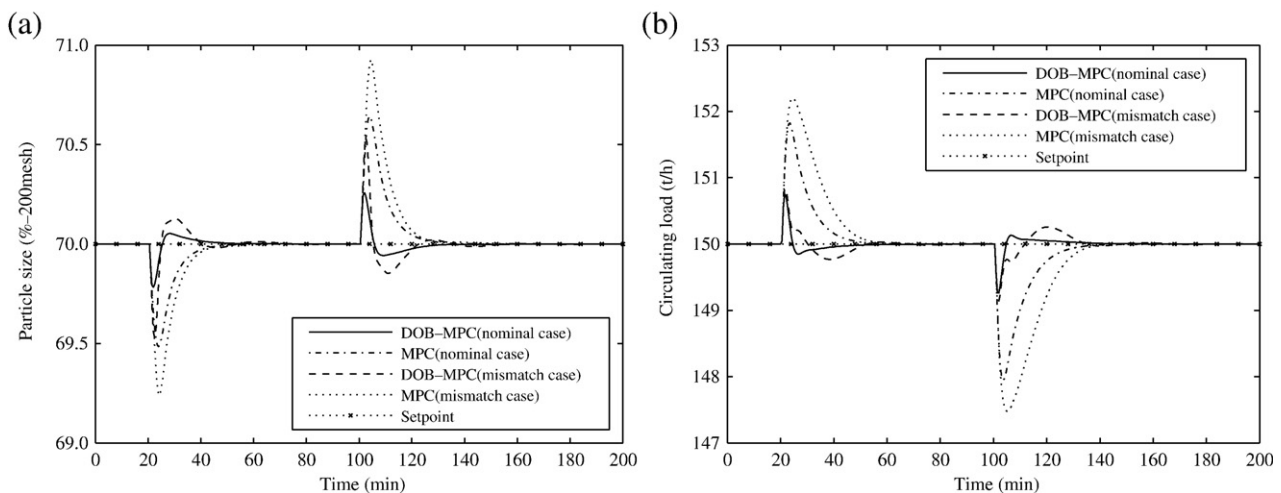


Fig. 6. Response curves of controlled variables in the presence of step external disturbances under DOB-MPC (solid line for nominal case and dashed line for model mismatch case) and MPC (dash-dot line for nominal case and dotted line for model mismatch case) schemes in both nominal case and model mismatch case: (a) particle size and (b) circulating load. The setpoints are denoted by x-marked dotted line.

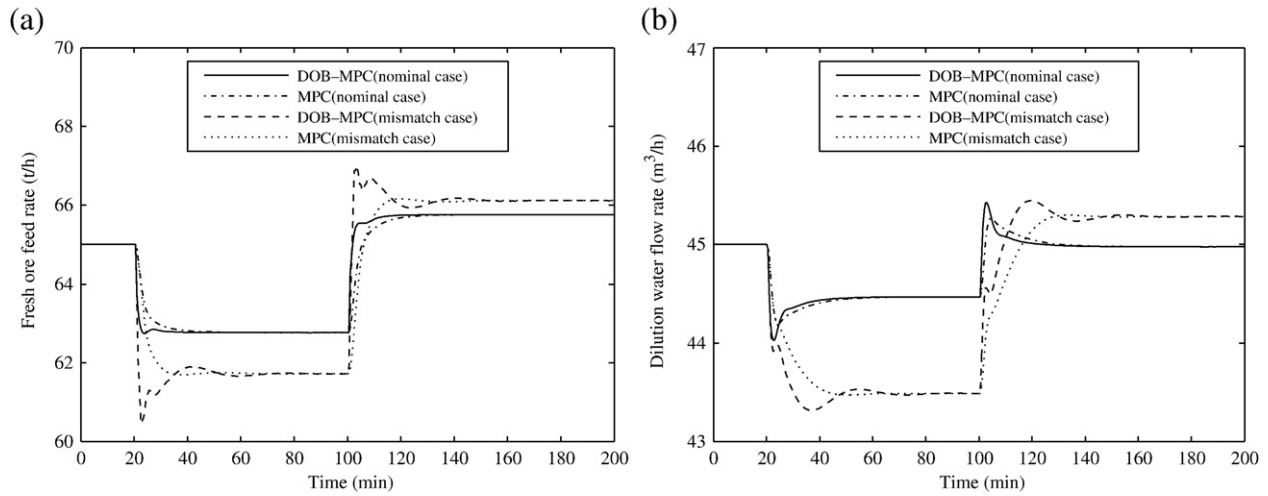


Fig. 7. Variations of manipulated variables in the presence of step external disturbances under DOB-MPC (solid line for nominal case and dashed line for model mismatch case) and MPC (dash-dot line for nominal case and dotted line for model mismatch case) schemes in both nominal case and model mismatch case: (a) fresh ore feed rate and (b) dilution water flow rate.

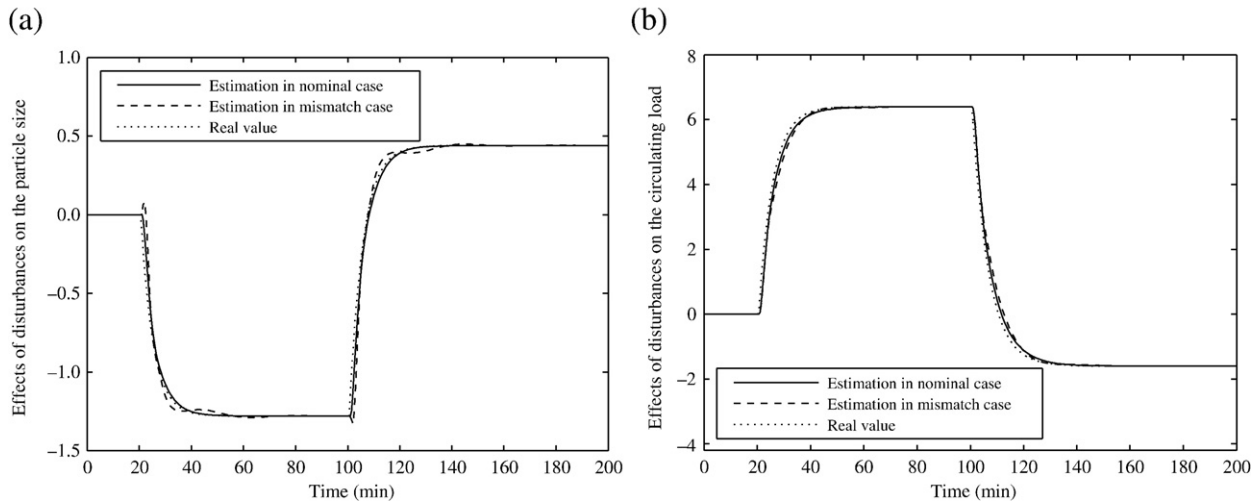


Fig. 8. Response curves of disturbances and their estimations in the presence of step external disturbance under DOB-MPC scheme in both nominal case (solid line) and model mismatch case (dashed line): (a) effects of the disturbances on the particle size and (b) effects of the disturbances on the circulating load. The real disturbances are denoted by dotted line.

that both the fluctuation amplitudes and frequencies of the two manipulated variables under the proposed method (solid line) are almost the same as those under the MPC scheme (dash-dot line). However, it can be observed from Fig. 10 that variations of the manipulated variables under the proposed method are a little quicker than those under the MPC method. Fig. 11 shows that the estimated (solid line) and the real (dotted line) disturbances almost overlap. Thus it can be said that the disturbance observer can estimate the disturbances effectively and the proposed method is more effective to overcome such sinusoidal external disturbance than the MPC scheme.

4.2. Disturbance rejection in the model mismatch case

In real practice, besides external disturbances, internal disturbance caused by model mismatches is another important factor which affects the control performance of the closed-loop systems. As illustrated in Section 3, the proposed method can not only reject external disturbances, but also the internal disturbance caused by model mismatches. In this part, we aim to demonstrate the lumped

disturbance rejection performance of the proposed method through simulation studies.

Suppose that the transfer function models of the process channels are expressed as

$$P_{11}(s) = \frac{-0.4}{3.5s + 1} e^{-0.5s}, \quad (32)$$

$$P_{12}(s) = \frac{3.42(1 - 0.994e^{-0.47s})}{(s + 1)(2s + 1)} e^{-0.13s}, \quad (33)$$

Table 2

Performance indexes in the presence of step external disturbances for the nominal case.

Performance index	Particle size		Circulating load	
	DOB-MPC	MPC	DOB-MPC	MPC
Overshoot(%)	0.37	0.92	0.50	1.37
IAE	2.77	9.80	7.12	37.97

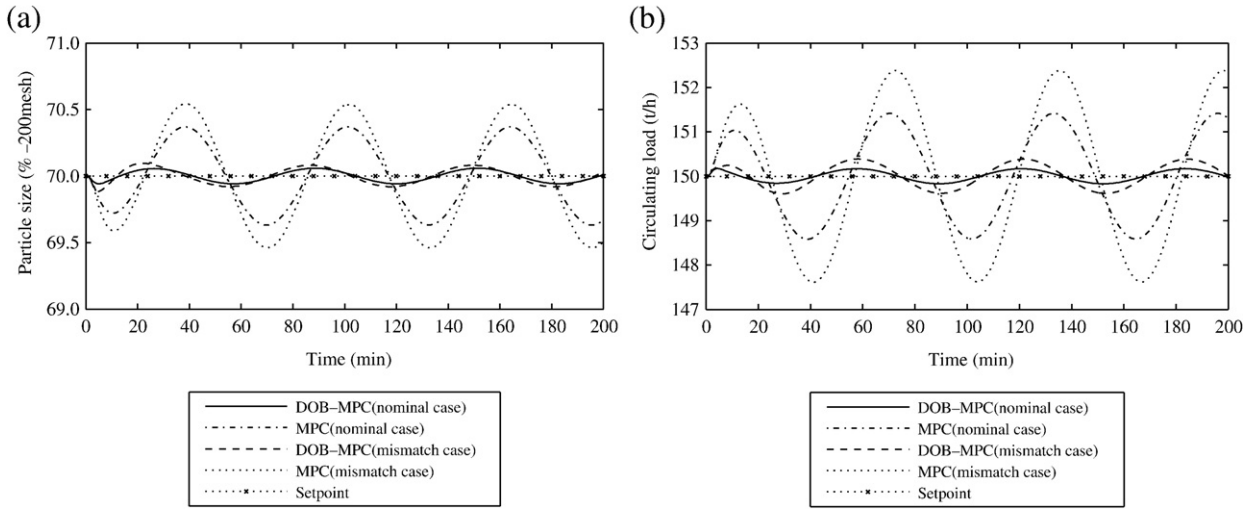


Fig. 9. Response curves of controlled variables in the presence of sinusoidal external disturbances under DOB-MPC (solid line for nominal case and dashed line for model mismatch case) and MPC (dash-dot line for nominal case and dotted line for model mismatch case) schemes in both nominal case and model mismatch case: (a) particle size and (b) circulating load. The setpoints are denoted by x-marked dotted line.

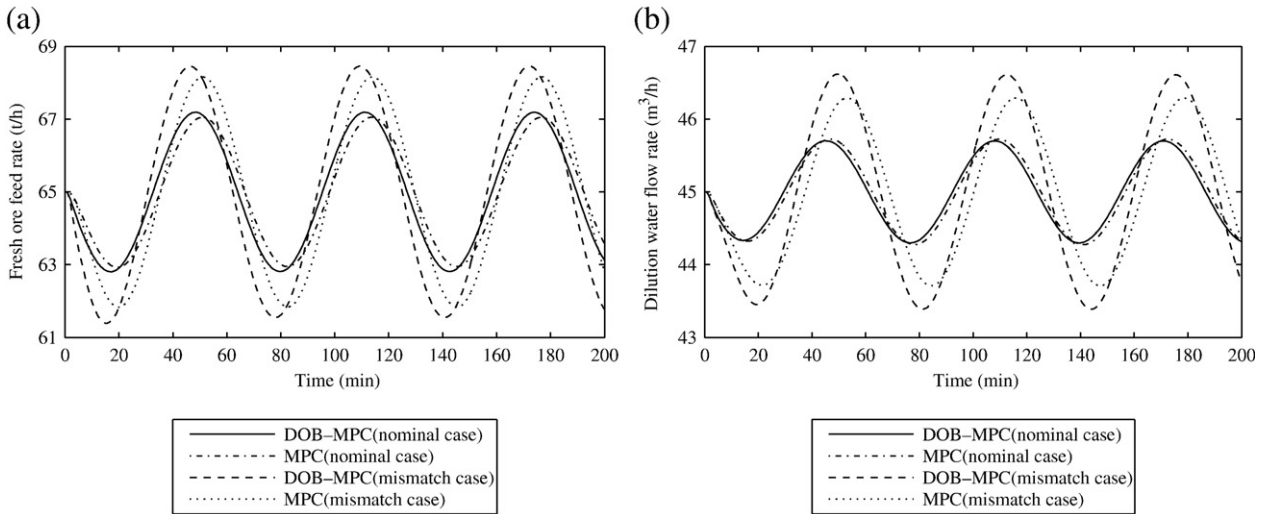


Fig. 10. Variations of manipulated variables in the presence of sinusoidal external disturbances under DOB-MPC (solid line for nominal case and dashed line for model mismatch case) and MPC (dash-dot line for nominal case and dotted line for model mismatch case) schemes in both nominal case and model mismatch case: (a) fresh ore feed rate and (b) dilution water flow rate.

$$P_{21}(s) = \frac{0.8}{3.2s + 1} e^{-0.3s}, \quad (34)$$

$$P_{22}(s) = \frac{2.5}{5.5s + 1} e^{-0.2s}. \quad (35)$$

Comparing Eqs. (24)–(27) with Eqs. (32)–(35), it can be found that severe model mismatches exist.

Case III. Step external disturbances in the model mismatch case: the ore hardness has an increase of 20% at $t=20$ min, while the feed particle size has a decrease of 20% at $t=100$ min.

The response curves of the controlled variables under the control of the two methods (dashed line for DOB-MPC and dotted line for MPC) in Case III are shown in Fig. 6. The corresponding response curves of the manipulated variables are shown in Fig. 7. Estimations of the disturbances in such case are shown in Fig. 8 (see curves in dashed line). Table 3 lists the performance indexes in such case.

Similar with Case I, it can be observed from Fig. 6 and Table 3 that the proposed method possesses smaller peak overshoot, smaller values of IAE and a faster convergence speed, etc. This means that the proposed method has achieved a much better step disturbance rejection performance than that of the MPC method even in the case of severe model mismatches. Fig. 8 shows that the error between the estimated and the real disturbances are also very small in such case.

Case IV. Sinusoidal external disturbances in the model mismatch case: variation of the ore hardness is sinusoidal, i.e., $d_{oh} = A \sin(\omega t)$, here $A = 20\%$, $\omega = 0.1$ rad/min.

In Case IV, the response curves of the controlled variables under the control of the proposed method (dashed line) and the MPC method (dotted line) are shown in Fig. 9. The corresponding changes of the manipulated variables are shown in Fig. 10. Fig. 11 shows the effects of estimated disturbances in such case (dashed line).

It can be seen from Fig. 9 that the fluctuation amplitudes of both the product particle size and the circulating load under the control of

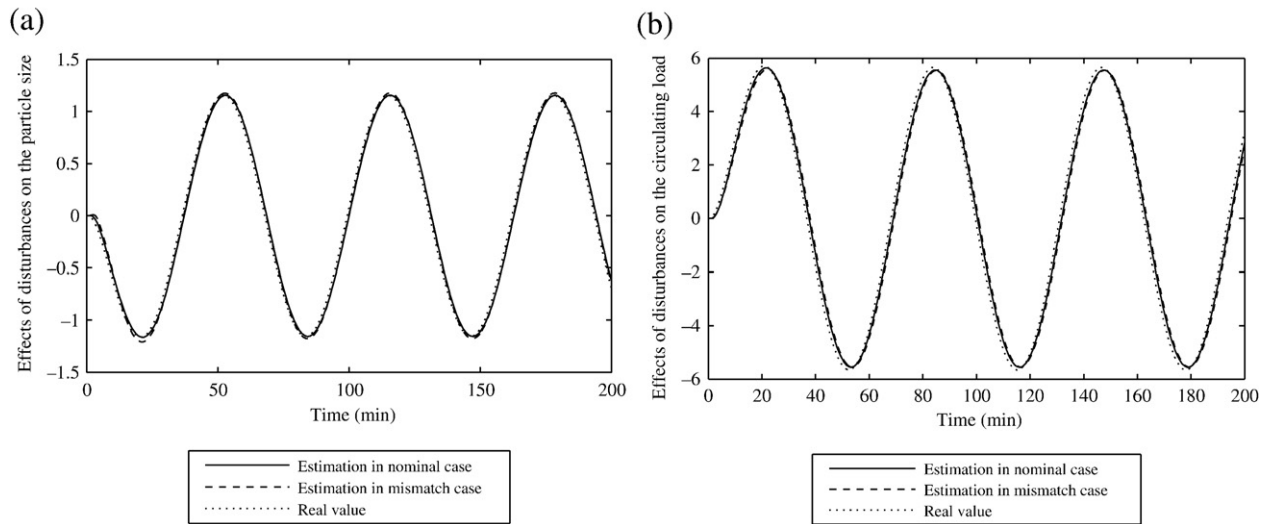


Fig. 11. Response curves of disturbances and their estimations in the presence of sinusoidal external disturbance under DOB-MPC scheme in both nominal case (solid line) and model mismatch case (dashed line): (a) effects of the disturbances on the particle size and (b) effects of the disturbances on the circulating load. The real disturbances are denoted by dotted line.

the proposed method (dashed line) are much smaller than those under the MPC method (dotted line). The fluctuation frequencies between the two methods are almost the same. Fig. 10 shows that the fluctuation amplitudes of the manipulated variables under the proposed method (dashed line) are a little larger than those under the MPC scheme (dotted line), but their changes are prior to those under the MPC method. It can be observed from Fig. 11 that the disturbance estimation errors in Case IV (difference between dashed line and dotted line) are also very small. Simulation results of Case IV demonstrate that the proposed method has remarkable superiorities in rejecting such lumped disturbances consisting of sinusoidal external disturbances and internal disturbances caused by model mismatches.

5. Conclusions

In grinding process, complex and unmeasurable disturbances generally have undesirable influences on the closed-loop system. Many existing methods including MPC demonstrate some limitations in the presence of strong disturbances. To improve the disturbance rejection property, a technique called disturbance observer (DOB) has been introduced for feedforward compensation design in this paper. A compound control scheme consisting of a feedforward compensation part based on DOB and a feedback regulation part based on MPC (DOB-MPC) has been developed. Besides external disturbances, the internal disturbance caused by model mismatches is merged into the terms of disturbance and is regarded as a part of the lumped disturbance. Rigorous disturbance rejection property analysis has also been given with the considerations of both model mismatches and external disturbances. Simulation results show that, compared with the control effects of the MPC method, the proposed method has

obtained remarkable superiority in rejecting the lumped disturbances in grinding circuits.

Nomenclature

A	dynamic matrix
e	vector of predicted errors
H	corrected coefficient matrix
m	number of the manipulated variables
M	control horizon
p	number of the controlled variables
P	prediction horizon
Q	error weighting matrix
R	input weighting matrix
$\tilde{y}_0$	the initial predicted output
$\tilde{y}_N$	the model prediction
$\tilde{y}_P$	the predicted output after correction
$\tilde{y}_r$	setpoint vector of the controlled variables

Greek symbols

Δu_M	control action of m manipulated variables in the next M sampling instance
λ_i	the time constant of the filter in the i th DOB
τ_{ij}	the time delay of $P_{ij}(s)$
τ_{ijn}	the time delay of $G_{ij}(s)$

Abbreviations

DOB	Disturbance observer
DOB-MPC	A compound scheme based on disturbance observer and model predictive control
FOPDT	first-order plus dead-time
IAE	Integral of absolute error
MIMO	Multi-input-multi-output
MPC	Model predictive control
RGA	Relative gain matrix

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Table 3
Performance indexes in the presence of step external disturbances for the model mismatch case.

Performance index	Particle size		Circulating load	
	DOB-MPC	MPC	DOB-MPC	MPC
Overshoot(%)	0.77	1.32	0.57	1.68
IAE	5.99	14.26	13.82	68.45

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